Grokking the System Design Interview

Learn how to prepare for system design interviews and practice common system design interview questions.

Level: None | Study Time: 21h | Lessons: 64

213,479 (Students)

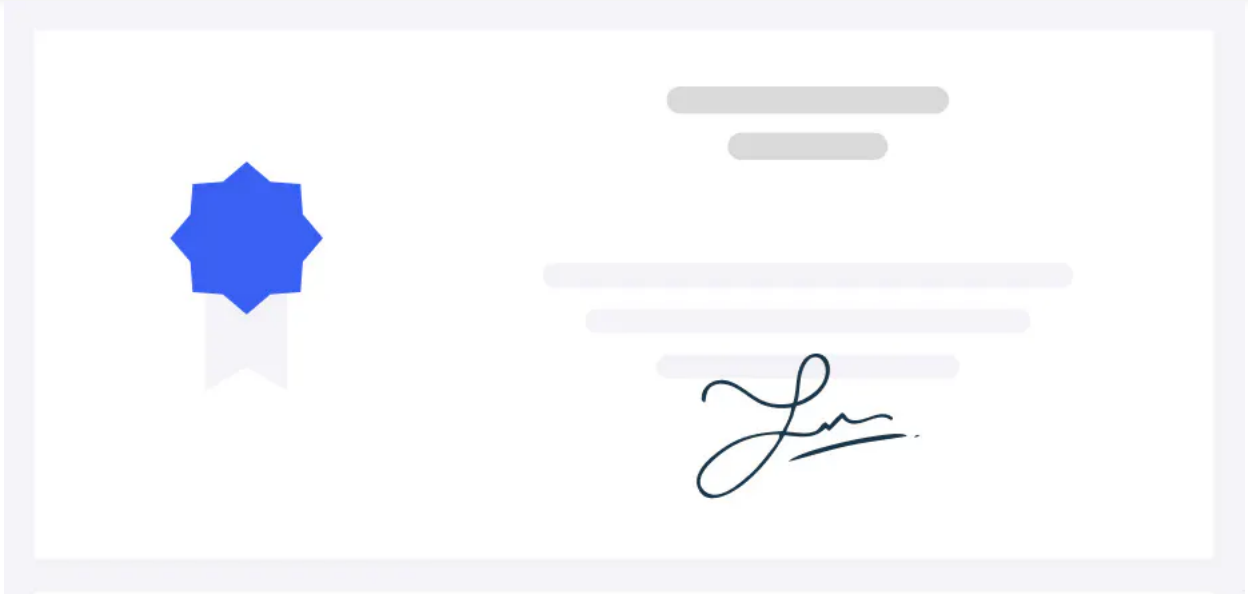
Course Overview

System design questions have become a standard part of the software engineering interview process. Performance in these interviews reflects upon your ability to work with complex systems and translates into the position and salary the interviewing company offers you. Unfortunately, most engineers struggle with the system design interview (SDI), partly because of their lack of experience in developing large-scale systems and partly because of the unstructured nature of SDIs. Even engineers who've some experience building such systems aren't comfortable with these interviews, mainly due to the open-ended nature of design problems that don't have a standard answer.

Grokking the System Design Interview is a complete guide to master the System Design Interview. It is created by Design Gurus who are a team of hiring managers from Google, Facebook, Microsoft, and Amazon. This course contains a carefully chosen set of questions that have not only been repeatedly asked at top tech companies but also provide a thorough experience to handle any system design problem.



Start Reading



Certification on completing the course.

Download Certificate

Course Content

- What is a System Design Interview?
- Functional vs. Non-functional Requirements
- What are Back-of-the-Envelope Estimations?
- Things to Avoid During System Design Interview

2. System Design Problems

18 chapters

- System Design Interviews - A step by step guide
- System Design Master Template
- Designing a URL Shortening Service like TinyURL
- Designing Pastebin
- Designing Instagram
- Designing Dropbox
- Designing Facebook Messenger
- Designing Twitter
- Designing Youtube or Netflix
- Designing Typeahead Suggestion
- Designing an API Rate Limiter
- Designing Twitter Search
- Designing a Web Crawler
- Designing Facebook's Newsfeed
- Designing Yelp or Nearby Friends
- Designing Uber backend
- Designing Ticketmaster
- Additional Resources

3. Glossary of System Design Basics

18 chapters

- System Design Basics
- Key Characteristics of Distributed Systems
- Load Balancing
- Caching
- Data Partitioning
- Indexes
- Proxies
- Redundancy and Replication
- SQL vs. NoSQL

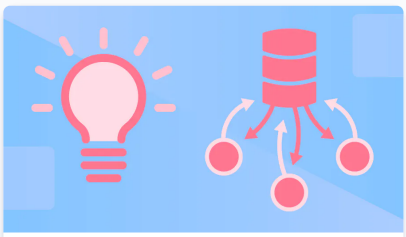
PACELC Theorem (New)
Consistent Hashing (New)
Long-Polling vs WebSockets vs Server-Sent Events
Bloom Filters (New)
Quorum (New)
Leader and Follower (New)
Heartbeat (New)
Checksum (New)

4. System Design Trade-offs — 22 chapters

Importance of Discussing Trade-offs
Strong vs Eventual Consistency
Latency vs Throughput
ACID vs BASE Properties in Databases
Read-Through vs Write-Through Cache
Batch Processing vs Stream Processing
Load Balancer vs. API Gateway
API Gateway vs Direct Service Exposure
Proxy vs. Reverse Proxy
API Gateway vs. Reverse Proxy
SQL vs. NoSQL
Primary-Replica vs Peer-to-Peer Replication
Data Compression vs Data Deduplication
Server-Side Caching vs Client-Side Caching
REST vs RPC
Polling vs Long-Polling vs Webhooks
CDN Usage vs Direct Server Serving
Serverless Architecture vs Traditional Server-based
Stateful vs Stateless Architecture
Hybrid Cloud Storage vs All-Cloud Storage
Token Bucket vs Leaky Bucket
Read Heavy vs Write Heavy System

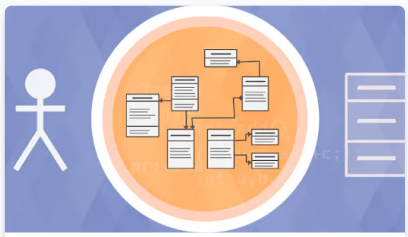
5. Appendix — 2 chapters

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Other courses



Grokking the Advanced System Design Interview

Start Reading



Grokking the Object Oriented Design Interview

Start Reading

What people say about our courses



Arijeet
Software Engineer

Just completed the “Grokking the system design interview”. It's amazing and super informative. Have come across very few courses that are as good as this!



Eric
Software Engineer

I've completed my first pass of "grokking the System Design Interview" and I can say this was an excellent use of money and time. I've grown as a developer and now know the secrets of how to build these really giant internet systems.



Brandon Lyons
Software Engineer

The famous "grokking the system design interview course" on http://designgurus.io is amazing. I used this for my MSFT interviews and I was told I nailed it.

MO JAFRI
Software Engineer

The courses which have "grokking" before them, are exceptionally well put together! These courses magically condense 3 years of CS in short bite-size courses and lectures (I have tried System Design, OODI, and Coding patterns). The Grokking courses are godsent, to be honest.

AHMET HANIF
Software Engineer

Whoever put this together, you folks are life savers. Thank you :)

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